Properties of equality and congruence

1

a $b + c$ **b** bc **c** AB

2 The multiplication property of equality.

3

a The symmetric property of equality. **b** The addition property of equality.

c The transitive property of congruence.

4

a No **b** No **c** Yes **d** No

5

a Yes **b** Yes **c** No **d** Yes

6

a No **b** No **c** No **d** Yes

7

a The transitive property of equality. **b** The multiplication property of equality.

c The addition property of equality.

8

a No **b** Yes **c** No **d** No

e Yes **f** No

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Additional questions

9

Property of Equality	Example
Reflexive property of equality	$a = a$
Symmetric property of equality	If $a = b$, then $b = a$.
Transitive property of equality	If $a = b$ and $b = c$, then $a = c$.
Addition property of equality	If $a = b$, then $a + c = b + c$.
Subtraction property of equality	If $a = b$, then $a - c = b - c$.
Multiplication property of equality	If $a = b$, then $a \times c = b \times c$.
Division property of equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.
Substitution property of equality	If $a =$ b, then b may be substituted for a in any expression containing a.

10

a Addition property of equality

b Converse of multiplication property

c Converse of addition property of equality

d Converse of subtraction property of equality

11

a Converse of transitive property of equality

b Substitution property of equality

© ClipSAT: By Mr. Mohamed Abdallah +966597688647